Photoelectric Effect Lab Report

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1 Introduction

The photoelectric effect is a phenomenon in quantum physics that demonstrates the quantized nature of light. Early experiments showed that electrons are emitted from a metal surface only when the frequency of incident light exceeds a threshold value, and that the kinetic energy of these emitted electrons only depend the frequency of the light, rather than its intensity as previously thought [1]. These observations cannot be explained by anything from classical physics, which would predict that energy transfer would be dependent on light intensity.

This discrepancy got resolved when Einstein proposed that light consists of photons, each with energy given by $E = hf$, where h is Planck's constant and f is the frequency of the light [1]. In his model, an electron is emitted from the metal if the photon energy exceeds the work function E_0 of the material, where the work function is defined as the minimum energy required to release an electron from the surface. The maximum kinetic energy of the emitted electrons can be measured experimentally using a stopping voltage, which is the potential required to reduce the photocurrent to zero.

In this set of experiments, the stopping voltage is measured for light of different frequencies using LEDs at different wavelengths. Planck's constant and the work function of the material that we are investigating are determined. From the linear fit to the stopping voltage compared to frequency, Planck's constant was empirically determined to be $(4.8 \pm 0.1) \times 10^{-34}$ J·s, and the work function was found to be (0.95 ± 0.04) eV. The corresponding threshold frequency was calculated to be $(2.3 \pm 0.1) \times 10^{14}$ Hz. While the measured value of Planck's constant is somewhat lower than the accepted value, our intermediate results are relatively consistent with the predicted linear relationships.

2 Equipment and Methodology

2.1 Equipment

The experiment uses an LED light source, a phototube, a controller box, and measuring instruments, which can be seen in Figure 1. The LED driver provides light of different wavelengths and adjustable intensity. This light is directed into the phototube, where it strikes the cathode and causes electrons to be emitted. The controller box converts the resulting photocurrent into a measurable voltage and also applies a variable stopping voltage using a built-in potentiometer. Two multimeters are used to measure the photocurrent (as a voltage) and the stopping voltage.

2.2 Methodology

This set of experiments was conducted in the same way for all the provided LEDs. We first put each LED into the phototube and turned it on. A variable stopping voltage was then applied using the controller box. The stopping voltage was gradually increased until the measured photocurrent approached zero. We repeated this stopping voltage measurement 5 times in order to account for uncertainties.

In a second part of the experiment, the dependence of photocurrent and stopping voltage on light intensity was investigated using a single LED. The intensity of the LED was varied by adjusting the input voltage to the LED driver. For each intensity setting, the photocurrent was measured as the voltage across a known resistance value, and the stopping voltage was also recorded.

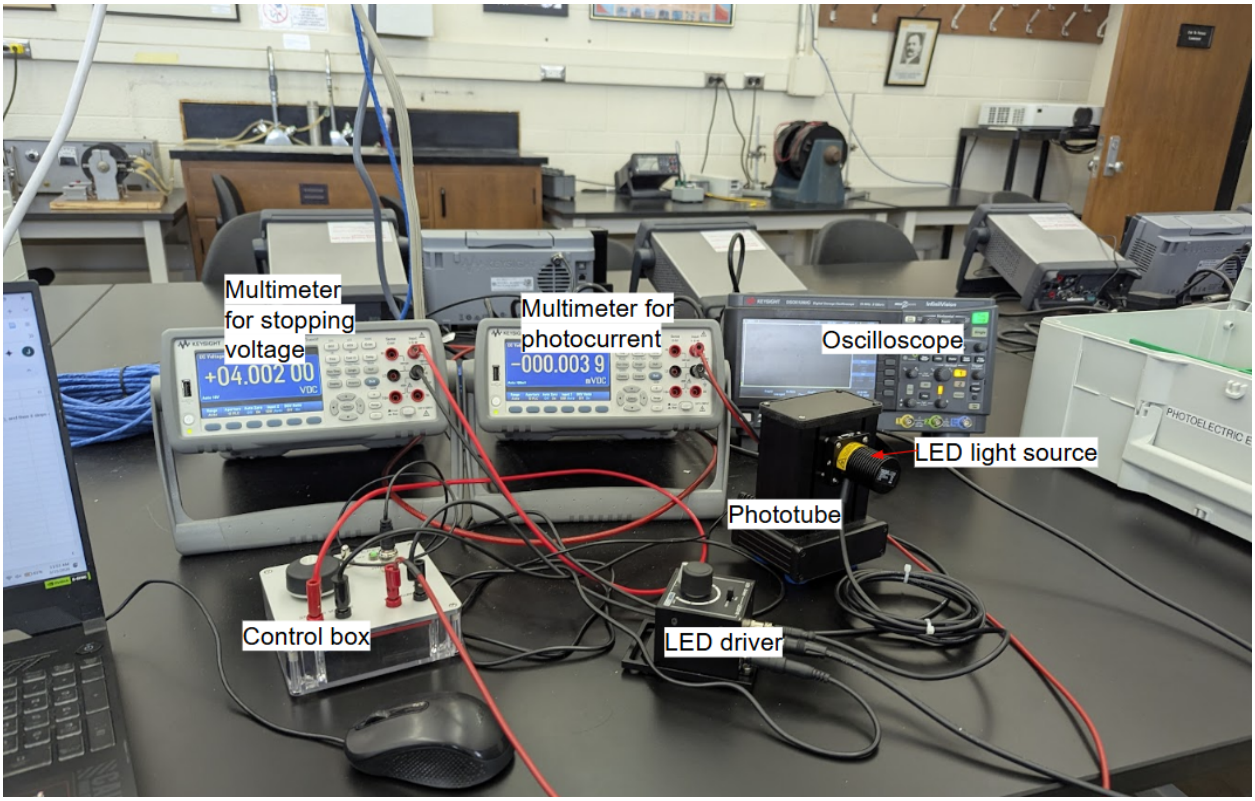

Figure 1: Our labelled experimental setup.

3 Results and Discussion

3.1 Calculating h, E_0 and f_0

We know that the energy of an ejected electron can be expressed in terms of the stopping voltage V_{stop}, the frequency of the photon that ejected it f, and the work function of the material E_0 using the equation:

$$eV_{stop} = hf - E_0 \tag{1}$$

We can rewrite this equation to extract a linear relationship for V_{stop} in terms of f, with a slope of $\frac{h}{e}$ and a y-intercept of $\frac{E_0}{e}$:

$$V_{stop} = \frac{h}{e}f - \frac{E_0}{e} \tag{2}$$

With this in mind, we plotted the data we collected relating the frequency of the light emitted by the LEDs and the resulting stopping voltage, which can be seen below:
The reduced chi-squared value is 1.899, implying a slight amount of underfitting. While the stopping voltage residual plots may imply a larger degree of underfitting, the extremely large frequency uncertainties lend credence to the idea that there is only a slight amount of underfitting. More information on the contribution of the frequency uncertainty to the reduced chi-squared value can be found in section (whatever uncertainty or chi sq section I throw this in)

Reading from the graph, the slope is $(3.01 \pm 0.08) \cdot 10^{-15}$Vs, so the corresponding value of Planck's constant can be found to be:

$$h = (3.01 \cdot 10^{-15})e \tag{3}$$

$$h = (3.01 \cdot 10^{-15})(1.602 \cdot 10^{-19}) \tag{4}$$

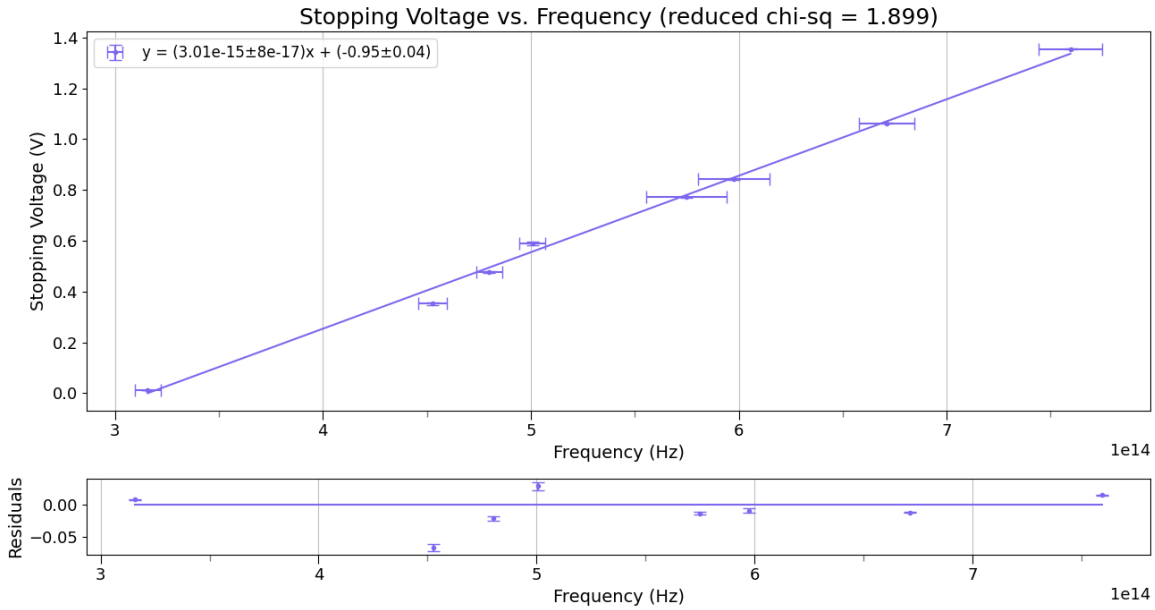

Figure 2: A graph showing the relationship between the frequency of light (in Hz) and the stopping voltage (in V).

$$h = (4.8 \pm 0.1) \cdot 10^{-34} \tag{5}$$

This does not fully agree with the literature value of Planck's constant, which is $6.6 \cdot 10^{-34}$ [2]. Some of this disagreement can be attributed to the extremely large bandwidths of the LEDs when driven with maximum current (as we did for this experiment). A larger bandwidth means a larger range of frequencies of light emitted by the LED, which in turn could result in a wider range of potential stopping voltages. Additionally, it could also be because of resistive losses in the measurement circuit, meaning the measured stopping voltage is actually less than the actual stopping voltage being output by the power supply, which in turn would decrease our slope and thus our experimental value of h. It could possibly be attributed to our variance in measuring the stopping voltage (since our stopping voltage was determined for extremely small measured values of photocurrent, but we were rarely ever able to find a stopping voltage with a zero value of photocurrent). However, it is also extremely likely that our inclusion of the M940L3 data point skewed our data and results in addition, which will be discussed in subsection 3.2

As for the values of E_0 and f_0, we can determine E_0 by noting that it is simply the product of the charge of an electron, e, and the negative of the y-intercept of the Voltage-Frequency graph above. Thus, we find:

$$E_0 = -(-0.95 \pm 0.04)e \tag{6}$$

$$E_0 = 0.95 \pm 0.04 \text{ eV} \tag{7}$$

Additionally, from E_0, we can calculate f_0 simply by dividing by h:

$$f_0 = \frac{0.95 \cdot (1.602 \cdot 10^{-19})}{6.63 \cdot 10^{-34}} \tag{8}$$

$$f_0 = (2.3 \pm 0.1) \cdot 10^{14} \text{ Hz} \tag{9}$$

Accordingly, the wavelength associated with this can be found as:

$$\lambda_0 = \frac{c}{f_0} \tag{10}$$

$$\lambda_0 = \frac{3.0 \cdot 10^8}{2.3 \cdot 10^{14}} \tag{11}$$

$$\lambda_0 = (1.30 \pm 0.06) \cdot 10^{-6} \text{ m} \tag{12}$$

While there is no reference value given for the work function or known datasheet for the phototube used in this experiment, we can still comment on some potential inconsistencies in our calculated values of E_0, f_0 and λ_0 and theory. Notably, if our calculated values were correct and theory is to be believed, then using LEDs with wavelengths smaller than λ_0 should yield nonzero stopping voltages. However, our stopping voltage for the M940L3, with a nominal peak wavelength of 950 nm, is essentially zero. Once again, a potential reason for this discrepancy can be attributed to our inclusion of the M940L3 data point, something which we will cover in the following subsection. While there is a possibility that there were resistive losses in the detection circuit that masked any potential photocurrent from the M940L3, when taken in conjunction with our chi-squared value of 1.899, it is absolutely possible that our work function

3.2 On the Inclusion of the M940L3 Data

As aforementioned, it is possible that our M940L3 data could be an outlier, as the work function of the phototube is such that 940 nm light is unable to generate a photocurrent. In this subsection, we argue that this may be a possibility, and that it can rectify some of the discrepancy between our results and theoretical predictions / literature. To begin, we consider the graph of our stopping voltage data without the M940L3 data:

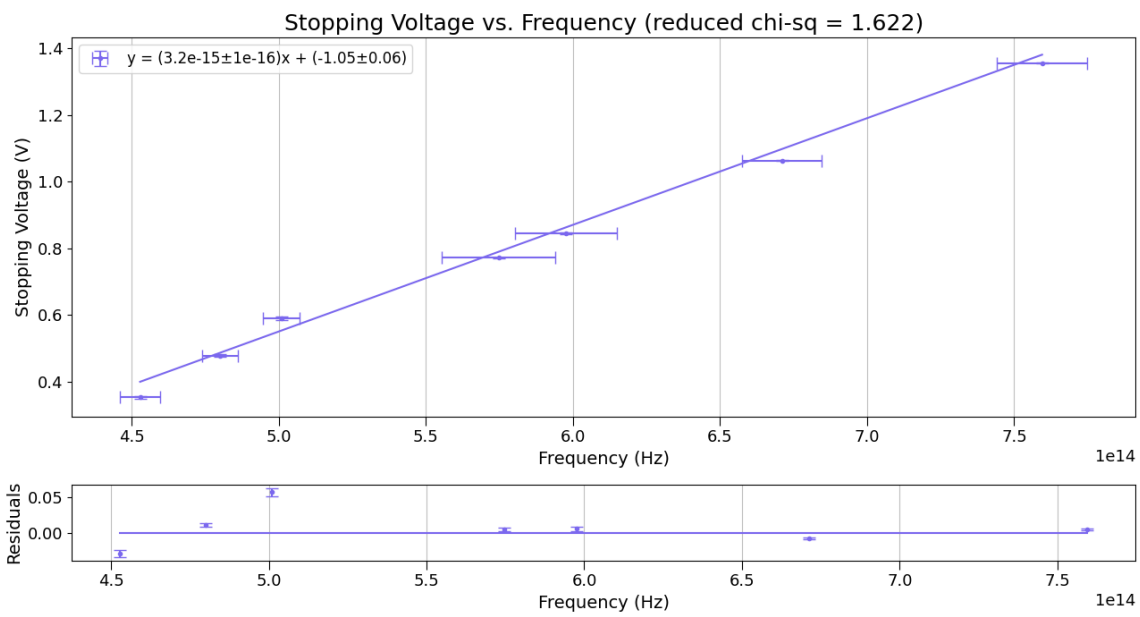

Figure 3: A graph showing the relationship between the frequency of light (in Hz) and the stopping voltage (in V), without the M940L3 data.

Immediately, we note that the reduced chi-squared value has decreased from 1.899 to 1.622, and that the variability in the residual plot has decreased as well, suggesting a better fit. Additionally, using the same methods detailed in the prior section, we note that the values for h, E_0, f_0 and λ_0 can be found to be:

$$h = (5.1 \pm 0.2) \cdot 10^{-34} \text{ Js} \tag{13}$$

$$E_0 = (1.05 \pm 0.06) \text{ eV} \tag{14}$$

$$f_0 = (2.5 \pm 0.1) \text{ Hz} \tag{15}$$

$$\lambda_0 = (1.18 \pm 0.03) \cdot 10^{-6} \text{ m} \tag{16}$$

The value of h is closer to the literature value provided of $6.6 \cdot 10^{-34}$ Js [2]. Though the value of E_0, f_0 and λ_0 are still theoretically inconsistent, they are closer to the values one would expect seeing that there was little to no stopping voltage associated with light with a wavelength of 950 nm. Though this is in no way a fully comprehensive argument for the exclusion of the M940L3 data, it does encourage further study using LEDs with nominal wavelengths between 950 and 660 nm, to better ascertain the true value of E_0 and f_0 of the phototube, and potentially resolve any doubt about whether or not the M940L3 data is an outlier.

3.3 The Relationship between Intensity, Photocurrent and Stopping Voltage

Since the intensity of the LED is directly proportional to the amount of current it is driven with, we can use the amount of current the LED is driven with as a proxy to investigate the relationship between the intensity, photocurrent and stopping voltage.

Since the LED driver was used in modulation mode for this experiment, we know from the LED driver datasheet that the amount of current sourced for a given voltage input simply equal to the value of the voltage divided by 5 [cite here]. With all of this in mind, we convert our initial voltage input (in V) into an LED current input (in A), and plot its relationship between the photocurrent (given by the voltage drop across the 100 MΩ resistor) [cite here], shown below.

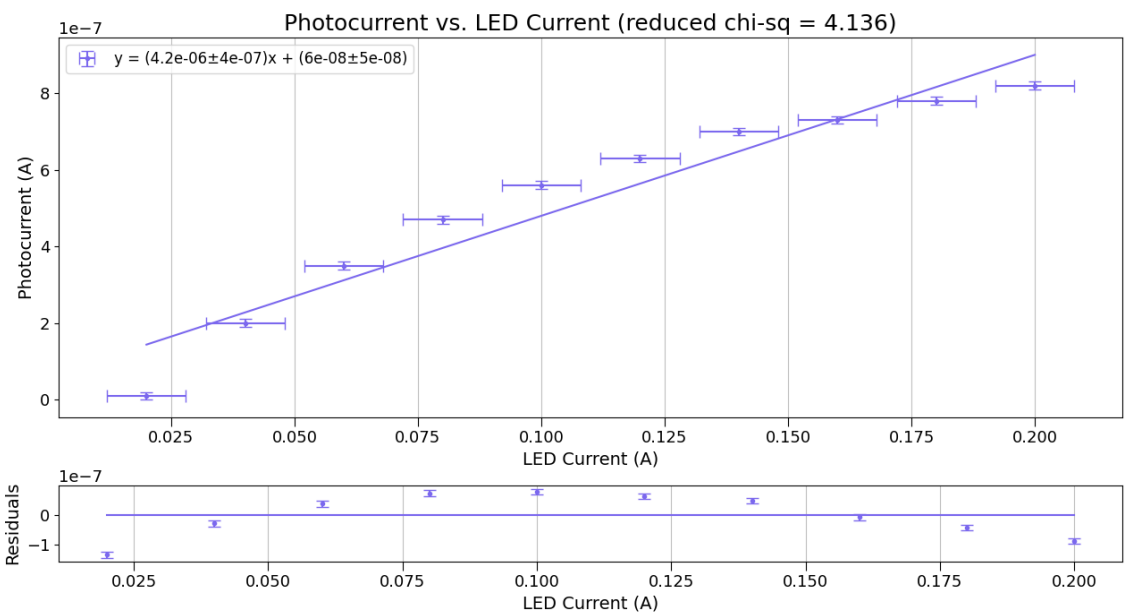

Figure 4: A graph showing the LED current input (in A) and the voltage produced by the resulting photocurrent (in V).

The slope of $(4.2 \pm 0.4) \cdot 10^{-6}$ supports the idea that there is a correlation between the LED current (or the intensity of light) and the photocurrent, which agrees with the literature result. However, the reduced chi-squared value is 4.136, implying there is some underfitting, and when taken in tandem with the shape of the residual plots, suggests a potential nonlinear correlation between the LED input current and the photocurrent. However, this nonlinearity may not be due to the nonlinearity of the LED at lower currents. While the datasheet shows a linear forward current / forward voltage graph even for very low currents [3], it

is still a possibility as the datasheet was likely written considering ideal operating conditions. Further investigation with higher LED input currents will likely be needed to determine whether or not this is truly the case.

In addition to considering the effect of intensity on photocurrent, one can also consider the effect of intensity on the stopping voltage by plotting the stopping voltages measured at a variety of LED input current values. The results can be seen in the plot below:

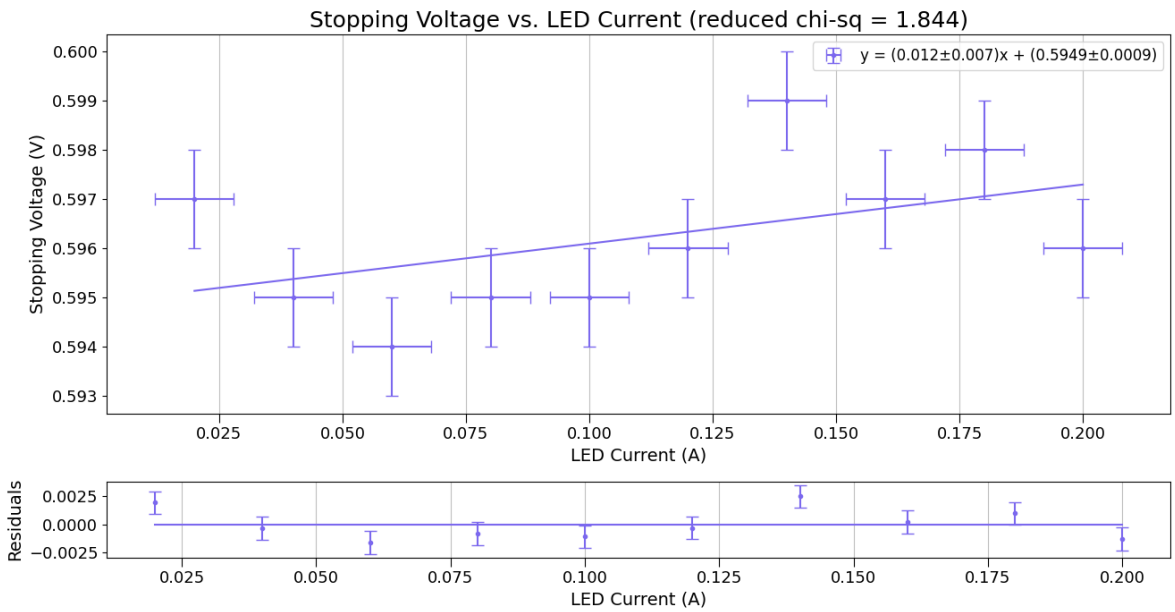

Figure 5: A graph showing the relationship between the LED input current (in A) and the stopping voltage (in V).

It is clear that there is little correlation between the LED current (or intensity of light) and the stopping voltage, with a slope of only 0.012 ± 0.007, barely larger than twice the uncertainty value. While the fact that a zero slope is not within the uncertainty might imply a small linear correlation, even this small amount of potential correlation is dubious, as the chi-squared value of 1.844 implies some amount of underfitting. Though the residual plot seems relatively balanced around the line, the actual position of the points on the scatter plot, and the large relative uncertainties in both LED current and stopping voltage both lend credence to the idea that the intensity of light has little effect on the stopping voltage, which agrees with the literature result as well.

4 Uncertainty Analysis

4.1 Sources and calculating uncertainty

In this experiment, measurement and instrumentation uncertainties were a significant contributing factor to our final uncertainty values. As such, we will detail the origin of those stated uncertainty values here.

4.2 Stopping Voltage

For almost all stopping voltage measurements, the largest source of uncertainty by far was from the variability in our measurements, not measurementnt error from our equipment. For each LED, we took five measurements of the stopping voltage, except for the M660L4, in which we took four, and took the sample

mean as our value of stopping voltage. This number is too small to invoke the Central Limit Theorem. However, we can assume that the population we are sampling (all possible stopping voltage measurements) is roughly normally distributed, since, as aforementioned, the largest source of uncertainty came from completely random variability (human error).

As a result, we can find our uncertainty value for our stopping voltage by calculating the sample standard deviation, and using a t-distribution with either 3 (for M660L4) or 4 (for everything else) degrees of freedom. Under this distrbution, t-values are related to the sample mean $\bar{V}$, population mean μ_V, sample standard deviation s and number of measurements n via the equation [4]:

$$t = \frac{\bar{V} - \mu_V}{s/\sqrt{n}} \tag{17}$$

To determine a value for our uncertainty using the t-distribution, we take the t-value associated with a 95% confidence interval, meaning that we are 95% confident that the true value of the stopping voltage lies within our uncertainty. Consulting the tables [5], we find that the t-value associated with a 95% confidence interval for 3 and 4 degrees of freedom are $t = 3.179$ and $t = 2.776$, respectively. Using these values and equation (17), we can determine the absolute uncertainty, ΔV, associated with variability in our stopping voltage measurements, and these are tabulated below (note that values of $\bar{V}$ and ΔV are given before rounding here):

Model	$\bar{V}$ (V)	s(V)	n	t-value	ΔV (V)
M940L3	0.012	0.0000187	5	2.776	0.0000232
M660L4	0.3516	0.00287	4	3.179	0.00456
M617L5	0.4778	0.00259	5	2.776	0.00321
M590L4	0.5904	0.00503	5	2.776	0.00624
M530L4	0.7708	0.00192	5	2.776	0.00239
M505L5	0.8444	0.00207	5	2.776	0.00257
M455L4	1.0634	0.000547	5	2.776	0.00068
M395L4	1.3546	0.000894	5	2.776	0.00111

Table 1: Tabulated values of $\bar{V}$, s, and ΔV from variability in our measurements, along with the t-values and values of n used to calculate ΔV.

For all our stopping voltage measurements, we considered a measurement uncertainty value of 0.001 V, as that was the significant digit that remained stable when taking our voltage measurements. Comparing this reading uncertainty value with the tabulated values of ΔV above, we see that all our ΔV values exceed our measurement uncertainty except for M904L3 and M455L4. As such, we take the larger of the two uncertainty values as our final uncertainty value, rounding everything to one significant digit, and tabulate the final uncertainty values below:

Model	$\bar{V}$ (V)	ΔV (V)
M940L3	0.012	0.001
M660L4	0.352	0.005
M617L5	0.478	0.003
M590L4	0.590	0.006
M530L4	0.771	0.002
M505L5	0.844	0.003
M455L4	1.063	0.001
M395L4	1.355	0.001

Table 2: Tabulated values of $\bar{V}$ (rounded), and ΔV considering the variability in our measurements and the measurement uncertainty of the stopping voltage

4.3 LED Wavelength and Frequency

To calculate the nominal LED wavelength and uncertainty in LED wavelength and frequency, we referenced the datasheet for each LED driver online. We calculated the uncertainty in wavelength as half the stated FWHM, and used the nominal peak LED wavelength as our nominal LED wavelength. This is justified as we always set the maximum LED current using the LED driver in its CW mode [reference the LED driver docs here], which should output a current value that is either at or near the peak current for all the LEDs. The values of our nominal LED wavelength and uncertainty in wavelength (unrounded) are tabulated below:

Model	Nominal Wavelength (λ) (nm)	FWHM (nm)	Uncertainty in Wavelength ($\Delta\lambda$) (nm)
M940L3 [[6]]	950	37	18.5
M660L4 [[7]]	662.5	20	10
M617L5 [[8]]	625	16	8
M590L4 [[9]]	599	15	7.5
M530L4 [[10]]	522	35	17.5
M505L5 [[11]]	502	29	14.5
M455L4 [[12]]	447	18	9
M395L4 [[3]]	395	16	8

Table 3: Nominal wavelengths and spectral bandwidths (FWHM) of the LEDs used in this experiment.

The propagation of this uncertainty for our frequency measurements will be discussed in later subsections (maybe add more to this if we have time).

4.4 Calculating Reduced Chi-Squared

Due to the significant amount of uncertainty present in the frequency measurements (especially when compared to the amount of uncertainty in stopping voltage measurements), it would be disingenous to only consider the uncertainty in stopping voltage measurements when calculating the reduced chi-squared value. To do this, instead of considering only the variance of y, we consider the effective variance σ_{eff} : [13]

$$\sigma_{eff} = \sigma_y^2 + (\frac{dy}{dx})^2 \sigma_x^2 \tag{18}$$

$$\sigma_{eff} = \sigma_y^2 + m^2 \sigma_x^2 \tag{19}$$

Thus, our modified reduced chi-squared value can be calculated using the formula:

$$\chi^2 = \sum_i^n \frac{(y_i - \bar{y}_i)^2}{\sigma_{ieff}^2} \tag{20}$$

Where $\bar{y}_i$ is the predicted value of y given by the linear relation:

$$\bar{y}_i = mx_i + b \tag{21}$$

4.5 Uncertainty Propagation for Constants

All of the constants in this lab were products of our uncertain values, so all uncertainty propagation necessary for these constants can be represented by the formula [14]:

$$\Delta z = z \sqrt{\sum_{i=0}^{n} (\frac{\Delta x_i}{x_i})^2} \tag{22}$$

Where z is a product or quotient of $x_1, x_2, \cdots, x_n$. The uncertainties are represented with the terms Δz and Δx_i.

4.6 Uncertainty Propagation for Linear and Proportional Fits

For a linear fit given by the equation $y = mx + b$ (in the case of a proportional fit, $b = 0$), the uncertainties for the slope m and y-intercept b are given by the equations [14]:

$$s_m = \sqrt{N\frac{s_{y,x}^2}{\Delta}} = \sqrt{\frac{N}{N-2}\frac{\sum[y_i - (mx_i + b)]^2}{N\sum x_i^2 - (\sum x_i)^2}} \tag{23}$$

$$s_b = \sqrt{\frac{s_{y,x}^2 \sum x_i^2}{\Delta}} = \sqrt{\frac{1}{(N-2)}\frac{\sum[y_i - (mx_i + b)]^2 \sum x_i^2}{N\sum x_i^2 - (\sum x_i)^2}} \tag{24}$$

$$\Delta = N\sum x_i^2 - (\sum x_i)^2 \tag{25}$$

$$s_{y,x}^2 = \frac{1}{N-2}\sum[y_i - (mx_i + b)]^2 \tag{26}$$

4.7 Sample Calculations

4.7.1 Sample Calculations for Uncertainty of f

For this lab, f was calculated using the equation:

$$f = \frac{c}{\lambda} \tag{27}$$

With c being the speed of light, or $3.0 \cdot 10^8$ m/s. The uncertainty, Δf, can be calculated using the equation

$$\Delta f = f\sqrt{(\frac{\Delta\lambda}{\lambda})^2}$$

$$\Delta f = \frac{f\Delta\lambda}{\lambda}$$

We take a sample calculation for λ of 599 nm, or $5.99 \cdot 10^{-7}$ m, with an associated f value of $5.008 \cdot 10^{14}$ Hz and a $\Delta\lambda$ value of 7.5 nm [9]. We then calculate Δf:

$$\Delta f = \frac{(5.008 \cdot 10^{14})(7.5 \cdot 10^{-9})}{(5.99 \cdot 10^{-7})}$$

$$\Delta f = 6.27 \cdot 10^{12} \text{ Hz} \tag{28}$$

$$\Delta f = 6 \cdot 10^{12} \text{ Hz} \tag{29}$$

4.7.2 Sample Calculations for Uncertainty in Slope

We perform sample calculations for the uncertainty in slope for the relationship between the stopping voltage and frequency.

$$s_m = \sqrt{\frac{8}{6}\frac{[(0.012 - (3.2 \cdot 10^{-15} \cdot (3.15 \cdot 10^{14})) - 1.05)]^2 + \cdots + [(1.355 - (3.2 \cdot 10^{-15} \cdot (7.59 \cdot 10^{14})) - 1.05)]^2}{8((3.15 \cdot 10^{14})^2 + \cdots + (7.59 \cdot 10^{14})^2) - (3.15 \cdot 10^{14} + \cdots + 7.59 \cdot 10^{14})^2}}$$

$$s_m = 1 \cdot 10^{-6}$$

4.7.3 Sample Calculations for Reduced Chi-Squared

Here, we perform sample calculations for our modified reduced chi-squared value for the relationship between the stopping voltage and frequency:

$$\chi^2 = \frac{([(0.012 - (3.2 \cdot 10^{-15} \cdot (3.15 \cdot 10^{14})) - 1.05)]^2}{\sigma^2_{0eff}} + \cdots + \frac{[(1.355 - (3.2 \cdot 10^{-15} \cdot (7.59 \cdot 10^{14})) - 1.05)]^2))}{\sigma^2_{8eff}} \tag{30}$$

$$\chi^2 = \frac{([(0.012 - (3.2 \cdot 10^{-15} \cdot (3.15 \cdot 10^{14})) - 1.05)]^2}{(0.001^2 + (3.2 \cdot 10^{-15})^2 (3.15 \cdot 10^{14})^2)} + \cdots + \frac{[(1.355 - (3.2 \cdot 10^{-15} \cdot (7.59 \cdot 10^{14})) - 1.05)]^2))}{(0.001^2 + (3.2 \cdot 10^{-15})^2 (7.59 \cdot 10^{14})^2)} \tag{31}$$

$$\chi^2 = 1.899 \tag{32}$$

4.8 Sources of error and uncertainty

One significant source of uncertainty came from the position of the LED relative to the phototube. In particular, the depth to which the LED was inserted into the phototube housing could not be held constant across LEDs. This would affect the effective intensity and spatial distribution of light incident on the cathode, which may cause some fluctuations in the photocurrent, which may then affect the stopping voltage.
Another major source of uncertainty is the fact that the LEDs have a finite spectral bandwidth. The labelled value is just a nominal peak of an (admittedly small) range of wavelengths. This introduces significant uncertainty in frequency and contributes directly to uncertainty in the slope of the linear fit. This effect is more pronounced at high current, when the peak would be broader. [15]

As well, the photocurrent does not abruptly drop to zero, but rather decreases gradually, making fairly difficult to identify a precise stopping point. The measurement is therefore somewhat subjective, which is why we did the analysis we did but is the reason for the larger uncertainty compared to other labs, where the human reading uncertainty may be less than the machine uncertainty. resulting in variability between repeated trials.

References

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